ABDURAHIMAN THOMMANGADAN

Senior AI Engineer | Generative AI, Multi-Agent Systems, LLMOps, Inference Engineering, RAG

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PROFESSIONAL SUMMARY

Senior AI Engineer with 7+ years of software engineering experience, leading production Generative AI since 2025. Technical Lead at Voizzit, where I architected CADBEATER from scratch: an AI estimator system for fenestration and facade projects that generates cost estimates, bills of quantities (BOQ), bills of materials (BOM) and CAD drawings from requirements, joinery details or floor plans. Hands-on across the modern AI stack: LangGraph multi-agent systems with MCP, OpenAI, Anthropic and OpenRouter LLMs, agent memory, agentic RAG, LoRA/PEFT fine-tuning, vLLM inference with quantization and constrained decoding, and Langfuse LLMOps, backed by strong system design and distributed-systems experience (Clean (Onion) Architecture, horizontal and vertical scaling, RBAC security).

TECHNICAL SKILLS

Generative AI & Agents: LangGraph, LangChain, Multi-Agent Systems, AI Agent Development, Agentic AI, Retrieval-Augmented Generation (RAG), Model Context Protocol (MCP), Tool Calling, ReAct, Reflexion, Plan-and-Execute, Human-in-the-Loop, Prompt Engineering, Agentic RAG, Qdrant, Vector Embeddings, Semantic Search

LLMs & Structured Decoding: OpenAI (GPT), Anthropic (Claude), OpenRouter, Open-Weight Models (Hugging Face), Structured Outputs, Constrained / Guided Decoding (JSON Schema)

Context & Memory: Context Engineering, Agent Memory (short-term and long-term), LangGraph Checkpointing, Context Compression, Token Budgeting

Inference Engineering: vLLM, Model Serving, Continuous Batching, KV Cache, Quantization (4-bit / 8-bit), Model Routing, Prompt & Response Caching, GPU Inference (RunPod), Token Streaming (SSE)

LLMOps & Evaluation: Langfuse, LLM Tracing, Prompt Versioning, Evaluation Pipelines, Regression Testing, Model Registry (Hugging Face), Model Versioning & Rollback, Cost & Latency Monitoring, Guardrails

Fine-Tuning & ML: LoRA, QLoRA, PEFT, Hugging Face Transformers, Vision-Language Models (VLM), Multimodal LLMs, Curriculum Learning, Dataset Curation, Deep Learning, Anomaly Detection

AI-Driven SDLC: Agentic Coding (Claude Code), Custom Module-Level & Project-Level Agent Skills, AI-Assisted Code Review, AI Development Lifecycle (data, fine-tuning, evaluation, deployment, monitoring)

Backend: Python, FastAPI, Django, Node.js, NestJS, Express.js, REST APIs, WebSocket, gRPC, ORM / ODM (Sequelize, TypeORM, Mongoose)

Architecture & System Design: Solution Architecture, System Design, Clean (Onion) Architecture, SOLID & Design Patterns, Test-Driven Development (TDD), Behavior-Driven Development (BDD), Microservices, Event-Driven Architecture (Kafka, RabbitMQ), Distributed Systems (MongoDB, PostgreSQL), Horizontal & Vertical Scaling, Caching (Redis), API Gateway, Asynchronous Workers & Queues, High Availability

Security: Role-Based Access Control (RBAC), JWT & OAuth 2.0 Authentication, OWASP Top 10, Secure API Design, Rate Limiting, Encryption in Transit & at Rest, Audit Logging

Databases & Messaging: MongoDB, PostgreSQL, Redis, Qdrant, Apache Kafka, RabbitMQ

Cloud & DevOps: AWS (EC2, S3, SQS), Docker, Kubernetes, GitHub Actions, CI/CD, ELK Stack

Frontend & Mobile: React.js, TypeScript, Three.js, Flutter

PROFESSIONAL EXPERIENCE

Voizzit Information Technology LLC, Dubai, UAE — Sep 2024 – Present

Technical Lead — Full Stack & Generative AI

Flagship project: CADBEATER

- Architected CADBEATER from requirements through production with a modular, horizontally scalable and secure design: an AI estimator system for fenestration and facade projects that generates cost estimates, bills of quantities (BOQ), bills of materials (BOM) and CAD drawings from requirements, joinery details or floor plans. The estimator reduced manual estimating effort from four hours to ten minutes (about 96% less time).
- Architected the LangGraph multi-agent system: a ReAct agent for contextual Q&A, a Reflexion agent with self-critique loops for vision-heavy requirement extraction, and a Plan-and-Execute agent for estimate generation behind mandatory human review, with tools and data sources integrated through Model Context Protocol (MCP).

- Integrated LLMs from OpenAI, Anthropic (Claude) and OpenRouter, routing each task between these APIs, the self-hosted fine-tuned model and deterministic code, with prompt and response caching.
- Engineered the inference layer: served the fine-tuned vision-language model with vLLM (continuous batching, KV cache) on RunPod GPUs, with 4-bit/8-bit quantization and structured outputs enforced through constrained (guided) decoding against JSON schemas.
- Fine-tuned a custom multimodal vision-language model with LoRA/PEFT on a three-stage curriculum (vision extraction, then pricing logic, then end-to-end drawing-to-estimate), with checkpoints versioned in a Hugging Face model registry for rollout and rollback.
- Built LLMOps on Langfuse: end-to-end tracing, prompt versioning, automated evaluation pipelines (test datasets, scoring, regression checks) gating every prompt and model release, and token-cost and latency monitoring per estimate.
- Designed agent context and memory management: short-term state persisted with LangGraph checkpointers so runs resume after human review, long-term memory of past projects and estimates, and context compression with per-node token budgets.
- Built agentic RAG on Qdrant and real-time token streaming (FastAPI, SSE, Redis Streams), and secured the platform with role-based access control (RBAC), role- and team-scoped retrieval, audit logging and human-in-the-loop approval gates.
- Drove an AI-driven SDLC for the team: agentic coding with Claude Code, custom agent skills for each project module plus shared project-level skills, and AI-assisted code review; authored the system architecture reference.

Senior Backend Engineer — SaaS Platforms

- Architected VData, a distributed voter-data platform on NestJS with Clean (Onion) Architecture and TypeORM, then scaled AWS search horizontally and vertically for fast, near-instant lookup across 30 million voter records.
- Built TechBuddy LMS (NestJS, PostgreSQL, React Native TV apps) with AI-generated quiz delivery.

Project: AutoFleet — Car Rental Booking & ERP Platform

- Built a car rental booking and ERP platform (booking flow, fleet, CRM and HRM modules) with a Flutter mobile app for Android, iOS and web, released and maintained on the Play Store and App Store.
  Tech stack: Node.js, React.js, Flutter (Dart), MongoDB, AWS

Samsung SDS India, Chennai, India — Feb 2024 – Sep 2024

Backend Engineer — Samsung.com Global eCommerce

- Built and operated independently deployable Node.js microservices for Samsung.com global eCommerce, supporting autoscaling during high-traffic sales events; used Kafka and RabbitMQ for event processing and PostgreSQL for transactional data.
- Built secure logistics and user-interaction APIs with Elasticsearch logging and monitoring.

Infosys Limited, Bangalore, India — Jun 2022 – Feb 2024

Associate Consultant — Enterprise SaaS

- Delivered high-availability microservices for Solargraf, a solar-design SaaS with 3D design and financial-calculation modules, behind an API Gateway with gRPC.

Kimball Electronics India Pvt. Ltd., Trivandrum, India — Nov 2020 – Jun 2022

Software Engineer — Industrial IoT

- Integrated machine-learning anomaly detection into real-time shop-floor sensor pipelines for EMTAB, an Industry 4.0 predictive-maintenance platform (Node.js, MongoDB), with Playwright test automation.

Nucore Software Solutions, Calicut, India — Apr 2019 – Nov 2020

Backend Developer — Travel Accounting SaaS

- Built the ERP accounting engine and microservice APIs for TRAACS, a back-office SaaS for travel agencies (Node.js, Express.js, PostgreSQL, MongoDB).

EDUCATION

B.Sc. Computer Science, Farook College, Kozhikode (University of Calicut) — 2016 – 2019

CERTIFICATIONS

- Machine Learning Specialization — DeepLearning.AI (Andrew Ng)
- Deep Learning — Codebasics